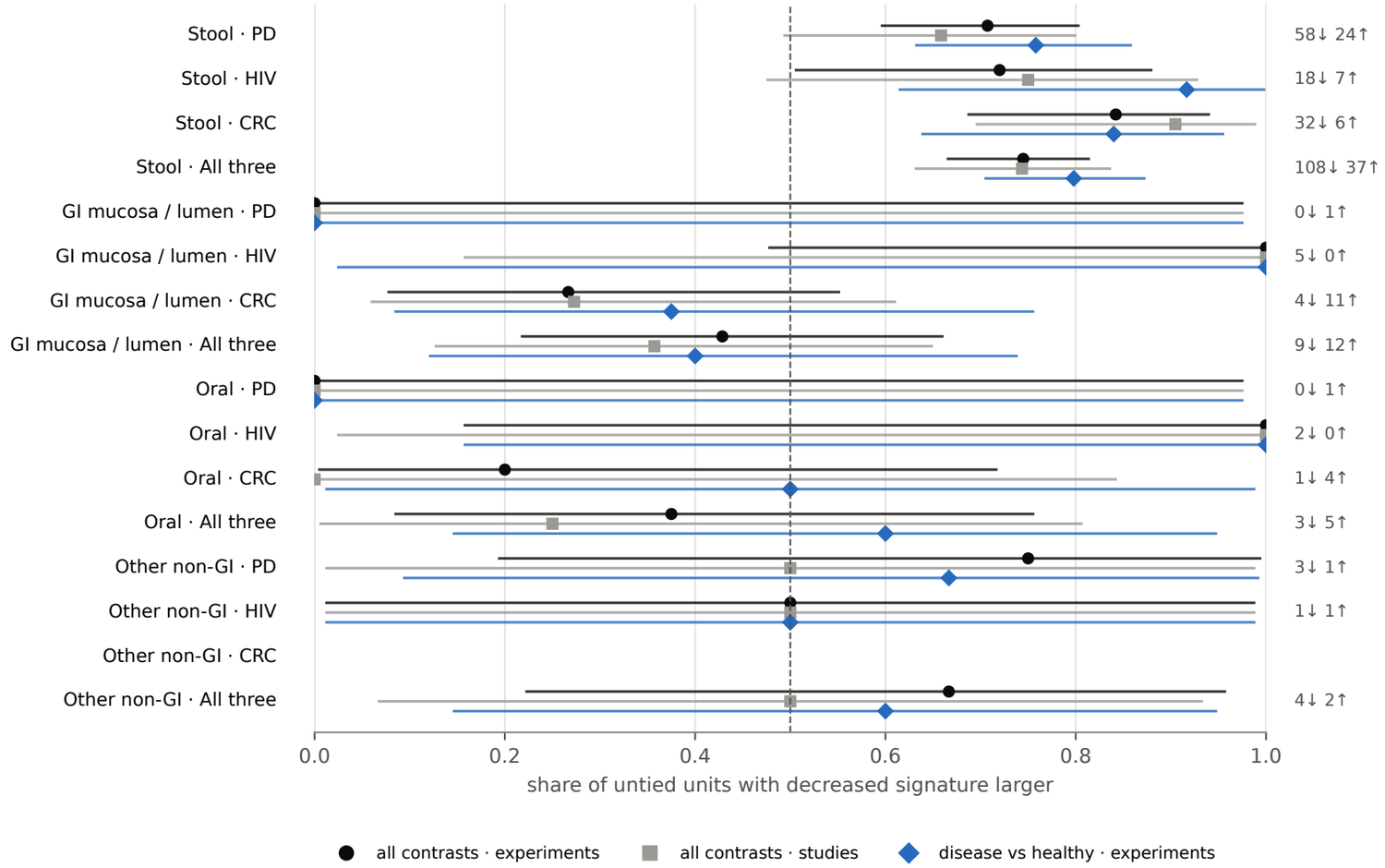
Direction of the cellulose annotation shift by body site, inference unit and contrast type

Share of untied units whose decreased signature had the larger cellulose annotation share, with exact 95% intervals; 0.5 means no preference.

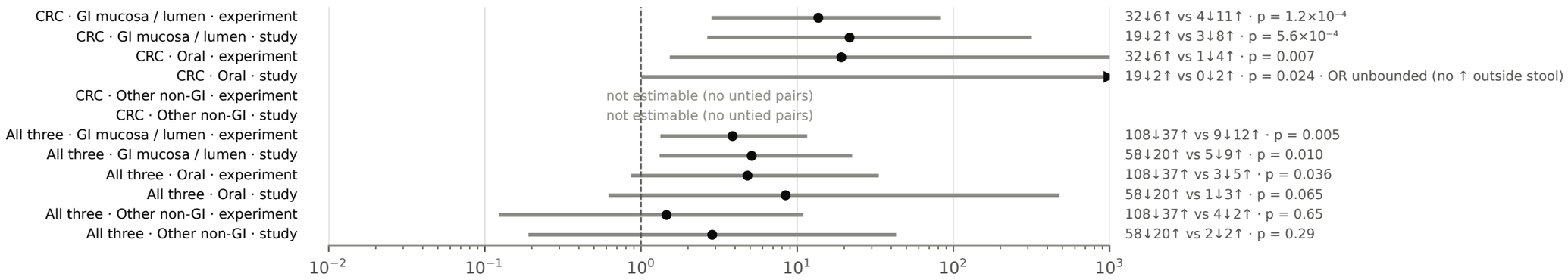
A. Share of untied units with the decreased signature larger



B. Set definitions: pairwise and mean d_J

	PD–HIV	PD–CRC	HIV–CRC	mean	core/union
Primary: all ranks	0.488	0.400	0.550	**0.479**	17/47
Genus and species ranks only	0.450	0.375	0.609	**0.478**	9/25
Profiles with cellulose z > 0 only	0.474	0.477	0.538	**0.496**	16/46
Disease-vs-healthy contrasts only	0.556	0.450	0.545	**0.517**	13/41
Disease-vs-healthy, strong evidence	0.556	0.450	0.545	**0.517**	13/41
All body sites, not only stool	0.551	0.446	0.588	**0.529**	19/60
Net decreased (more dec than inc profiles)	0.645	0.622	0.657	**0.641**	7/40
Net decreased (more dec than inc studies)	0.586	0.667	0.676	**0.643**	7/39
Baseline: all stool-decreased taxa, any trait	0.795	0.747	0.783	**0.775**	51/471

C. Stool against each other compartment (odds ratio of the ↓ share, exact 95% interval)



Experiments within a study are not independent; the study rows collapse them to one vote per study. Disease-vs-healthy rows keep experiments the contrast manifest classifies disease_vs_healthy, relabelled so decreased means lower in disease. Intervals are exact (Clopper–Pearson for shares; conditional maximum likelihood for odds ratios). PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.